DEPARTMENT OF MATHEMATICS

MIDTERM TEST #2

MTH 240 – Calculus II

Last Name (Print):_____. First Name:_____. Student Number: _____

Signature:_____.

Date: June 15, 2012, 4:00 pm

Duration: 90 minutes

Section (circle one)

Dr. B. Tasić :	1 2 3 7
Dr. A. Alvarez :	4 5 6 8

Instructions:

1. This is a closed-book test. **Notes, calculators and other aids are not permitted.**
2. Verify that your test has pages 1-7
3. (a) Unless otherwise instructed, **make sure you include all significant steps in your solution, presented in the correct order. Unjustified answers will be given little or no credit. Cross out or erase all rough work not relevant to your solution.** Put a box around your final **answer.**

   (b) Write your solutions in the space provided. If you need more space, use the back of the page. Indicate this fact on the original page, making sure that your solution cannot be confused with any rough work which may be there. Marks (out of 50) are shown in brackets.
4. Do not separate the sheets.
5. Have your student card available on your desk.

For Instructor's use only.

Page	Mark
2	/
3	/
4	/
5	/
6	/
7	/
Total	/50

1. [7 marks] Show that the series

$$\sum_{n=1}^{\infty}\left(\frac{1}{\sqrt{2n}}-\frac{1}{\sqrt{2n+2}}\right)$$

is convergent and find its sum.

Using Telescoping we have

$$S_n = \sum_{K=1}^{n}\left(\frac{1}{\sqrt{2K}}-\frac{1}{\sqrt{2K+2}}\right) = \sum_{K=1}^{n}\frac{1}{\sqrt{2K}} - \sum_{K=1}^{n}\frac{1}{\sqrt{2K+2}} =$$

$$= \left(\frac{1}{\sqrt{2}}+\cancel{\frac{1}{\sqrt{4}}}+\cdots+\cancel{\frac{1}{\sqrt{2n}}}\right)-\left(\cancel{\frac{1}{\sqrt{4}}}+\cancel{\frac{1}{\sqrt{6}}}+\cdots+\cancel{\frac{1}{\sqrt{2n}}}+\frac{1}{\sqrt{2n+2}}\right)$$

$$= \frac{1}{\sqrt{2}}-\frac{1}{\sqrt{2n+2}}.$$ Hence $\lim_{n\to\infty} S_n = \lim_{n\to\infty}\left(\frac{1}{\sqrt{2}}-\frac{1}{\sqrt{2n+2}}\right)=\frac{1}{\sqrt{2}}$

So, $\sum_{n=1}^{\infty}\left(\frac{1}{\sqrt{2n}}-\frac{1}{\sqrt{2n+2}}\right)=\frac{1}{\sqrt{2}}$, convergent.

2. [3 marks] Show that the series

$$\sum_{n=1}^{\infty}\cos\frac{n^2+1}{2n^2+1}$$

diverges.

Put $a_n = \cos\frac{n^2+1}{2n^2+1}$. Since $\lim_{n\to\infty} a_n = \lim_{n\to\infty}\cos\frac{n^2+1}{2n^2+1} =$

$= \cos\lim_{n\to\infty}\frac{n^2+1}{2n^2+1} = \cos\frac{1}{2} \neq 0$ by the Test for Divergence

we have that $\sum_{n=1}^{\infty}\cos\frac{n^2+1}{2n^2+1}$ is divergent.

3. [10 marks] Use the Integral Test to show that the series

$$1+\frac{4}{e}+\frac{9}{e^2}+\frac{16}{e^3}+\frac{25}{e^4}+\ldots\ldots\ldots$$

is convergent.

We see that $a_n = \frac{n^2}{e^{n-1}} = n^2 e^{1-n}$ and the above series is $\sum_{n=1}^{\infty} n^2 e^{1-n}$. By $\int$-TEST we have that $\sum_{n=1}^{\infty} n^2 e^{1-n}$ is conv iff $\int_1^{\infty} x^2 e^{1-x}\,dx$ is conv.

Put $f(x) = x^2 e^{1-x}$, $f(n) = a_n$ and consider $x \in [1,\infty)$

1) f is obviously positive and continuous function on $[1,\infty)$

2) $f'(x) = 2xe^{1-x} + x^2 e^{1-x}\cdot(-1) = xe^{1-x}(2-x) < 0$ iff $x > 2$. So $f \searrow$ on $[2,\infty)$

$\int_1^{\infty} x^2 e^{1-x}\,dx = \lim_{t\to\infty} \int_1^t x^2 e^{1-x}\,dx = 5$ and hence convergent

To compute $\int_1^t x^2 e^{1-x}\,dx$ we can use $\int$-by parts

$$\int_1^t x^2 e^{1-x}\,dx = e\int_1^t x^2 e^{-x}\,dx = e\left[-x^2 e^{-x}\Big|_1^t + 2\int_1^t x e^{-x}\,dx\right]$$

$x^2 = u$, $2x\,dx = du$, $e^{-x}dx = dv$, $v = -e^{-x}$ $\qquad$ $x = u$, $dx = du$, $e^{-x}dx = dv$, $v = -e^{-x}$

$$= e\left[-x^2 e^{-x}\Big|_1^t + 2\left(-xe^{-x}\Big|_1^t + \int_1^t e^{-x}\,dx\right)\right]$$

$$= e\left[-x^2 e^{-x} - 2xe^{-x} - 2e^{-x}\right]\Big|_1^t = -e^{1-x}\left[x^2+2x+2\right]\Big|_1^t$$

$$= -e^{1-t}(t^2+2t+2) + [1^2+2+2] = -e^{1-t}(t^2+2t+2)+5$$

Since $\lim_{t\to\infty} -e^{1-t}(t^2+2t+2) = -e\lim_{t\to\infty}\frac{t^2+2t+2}{e^t} \overset{\frac{\infty}{\infty}}{=} -e\lim_{t\to\infty}\frac{2t+2}{e^t} \overset{\frac{\infty}{\infty}}{=}$

$= -e\lim_{t\to\infty}\frac{2}{e^t} = 0$ (we used twice L'Hospital Rule)

we get $\lim_{t\to\infty}\left(-e^{1-t}(t^2+2t+2)+5\right) = 5$

4. [10 marks] Does the series

$$\sum_{n=2}^{\infty}(-1)^n\frac{\ln n}{\sqrt{n}}$$

converge absolutely, conditionally or diverge? Justify your answer.

Absolute Convergence: Consider $\sum_{n=2}^{\infty}\left|(-1)^n\frac{\ln n}{\sqrt{n}}\right| = \sum_{n=2}^{\infty}\frac{\ln n}{\sqrt{n}}$

Claim: $\sum_{n=1}^{\infty}\frac{\ln n}{\sqrt{n}}$ is divergent

Using Comparison Test: $a_n = \frac{\ln n}{\sqrt{n}} \geq \frac{1}{\sqrt{n}} = b_n$ for $n \geq 3$

Since $\sum_{n=2}^{\infty}\frac{1}{\sqrt{n}}$ is $p=\frac{1}{2}$-Harmonic divergent series we get that $\sum_{n=2}^{\infty}\frac{\ln n}{\sqrt{n}}$ is divergent by the Comparison Test.

[REMARK: You can use $\int$-TEST to show that $\sum_{n=2}^{\infty}\frac{\ln n}{\sqrt{n}}$ is DIV]

Convergence: $\sum_{n=2}^{\infty}(-1)^n\frac{\ln n}{\sqrt{n}} = \sum_{n=1}^{\infty}(-1)^n a_n$

Claim: $\sum_{n=2}^{\infty}(-1)^n\frac{\ln n}{\sqrt{n}}$ is convergent by the Alternating Series Test

(1) $a_n = \frac{\ln n}{\sqrt{n}} > 0$ for $n \geq 2$

(2) $\lim_{n\to\infty} a_n = \lim_{n\to\infty}\frac{\ln n}{\sqrt{n}} = 0$

$$\lim_{x\to\infty}\frac{\ln x}{\sqrt{x}} \underset{\infty}{\overset{\infty}{=}} \lim_{x\to\infty}\frac{\frac{1}{x}}{\frac{1}{2\sqrt{x}}} = \lim_{x\to\infty}\frac{2}{\sqrt{x}} = 0$$

(3) a_n is decreasing

$f(x) = \frac{\ln x}{\sqrt{x}}$, $f(n) = a_n$ $n \geq 2$

$f'(x) = \frac{2-\ln x}{2x^{3/2}} < 0 \Leftrightarrow \ln x > 2 \Leftrightarrow x > e^2$

So f is eventualy [from $[e^2, \infty)$ decreasing

By the alternating Series test $\sum_{n=2}^{\infty}(-1)^n\frac{\ln n}{\sqrt{n}}$ is conv

and hence it is conditionally convergent.

5. [6 marks] Determine whether the series

$$\sum_{n=1}^{\infty} \frac{1+3^n}{1+4^n}$$

is convergent or divergent. Justify your answer.

Using the Limit Comparison Test :

Put $a_n = \frac{1+3^n}{1+4^n}$ and $b_n = \frac{3^n}{4^n} = \left(\frac{3}{4}\right)^n$. Then $a_n, b_n > 0$

$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \left(\frac{3}{4}\right)^n$ is a convergent Geom Series $\left(a = \frac{3}{4} = r\right)$

Since $\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{1+3^n}{1+4^n}}{\frac{3^n}{4^n}} = \lim_{n\to\infty} \frac{4^n(1+3^n)}{3^n(1+4^n)} = \lim_{n\to\infty} \frac{4^n \cdot 3^n\left(\frac{1}{3^n}+1\right)}{3^n \cdot 4^n\left(\frac{1}{4^n}+1\right)}$

$= \lim_{n\to\infty} \frac{\cancel{\frac{1}{3^n}}^{\to 0} + 1}{\cancel{\frac{1}{4^n}}^{\to 0} + 1} = 1$

We get that $\sum_{n=1}^{\infty} \frac{1+3^n}{1+4^n}$ is also conv. by the L.C.T.

REMARK : You can also use Comparison Test and compare a_n with $b_n = 2 \cdot \frac{3^n}{4^n}$

6. [8 marks] Solve the differential equation

$$y' + y\cos x = \sin 2x.$$

METHOD 1: Put $y = u \cdot v$. Then $y' = u'v + u \cdot v'$ and the equation becomes

$$u' \cdot v + uv' + u \cdot v\cos x = \sin 2x$$
$$u' \cdot v + u(v' + v\cos x) = \sin 2x \quad \cdots \quad (*)$$

Now find v s.t $v' + v\cos x = 0 \iff \frac{dv}{v} = -\cos x\, dx \Rightarrow$
$\ln|v| = -\sin x \Rightarrow v = e^{-\sin x}$ (we need only one v!)

Using this v $(*)$ becomes $u' e^{-\sin x} = \sin 2x \iff u' = e^{\sin x} \sin 2x$
and thus $u = \int e^{\sin x} \sin 2x\, dx$

$$\int e^{\sin x} \sin 2x\, dx = \int e^{\sin x} 2\sin x \cos x\, dx = 2\int e^t \cdot t\, dt = \ldots$$

put $\sin x = t \Rightarrow \cos x\, dx = dt$

$$= 2e^t(t-1) + C = 2e^{\sin x}(\sin x - 1) + C$$

Finally $y = u \cdot v = \left(2e^{\sin x}(\sin x - 1) + C\right) \cdot e^{-\sin x}$

$$y = 2(\sin x - 1) + Ce^{-\sin x}$$

METHOD 2: The integrating factor $I(x) = e^{\int \cos x\, dx} = e^{\sin x}$

So $e^{\sin x} \cdot y' + y\cos x \cdot e^{\sin x} = e^{\sin x} \cdot \sin 2x$ becomes

$$(y e^{\sin x})' = e^{\sin x} \sin 2x \Rightarrow$$

$$y e^{\sin x} = \int e^{\sin x} \sin 2x\, dx = 2e^{\sin x}(\sin x - 1) + C$$

Hence

$$y = 2(\sin x - 1) + Ce^{-\sin x}$$

7. [6 marks] Solve the initial-value problem

$$y' = 3y - 2x + 6xy - 1, \qquad y(0) = 1.$$

$y' = 3y(2x+1) - (2x+1) = (2x+1)(3y-1)$. Hence

$\frac{dy}{3y-1} = (2x+1)\,dx$, $y \neq \frac{1}{3}$

$\int \frac{dy}{3y-1} = \int (2x+1)\,dx \Rightarrow \frac{1}{3}\ln|3y-1| = x^2 + x + C_1$

$\Rightarrow \ln|3y-1| = 3x^2 + 3x + 3C_1 \Rightarrow |3y-1| = e^{3C_1} \cdot e^{3x^2+3x}$

$\Rightarrow 3y - 1 = \pm e^{3C_1} e^{3x^2+3x} \Rightarrow y = \frac{1}{3} + Ce^{3x^2+3x}$, $C \in \mathbb{R}$

($y = \frac{1}{3}$ is also a solution of the equation)

$y(0) = \frac{1}{3} + Ce^0 \Rightarrow 1 = \frac{1}{3} + C \Rightarrow C = \frac{2}{3}$

So $y = \frac{1}{3} + \frac{2}{3}e^{3x^2+3x}$